

MANGAIA, Cook Islands

WMO: 918480

Lat: 21.917S

Lon: 157.950W

Elev: 41

StdP: 100.83

Time Zone: -10.00 (W10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	17.2	10.0	7.7	19.8	10.9	8.2	19.7	12.0	21.0	10.0	21.8	1.5	120	0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	29.9	25.2	29.2	25.1	28.7	24.9	26.5	28.5	26.1	28.1	25.7	27.7	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.5	28.0	25.4	20.7	27.5	25.1	20.3	27.1	83.3	28.5	81.2	28.3	79.4	27.8	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.1	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	25.7	26.4	26.1	24.9	23.6	22.4	21.8	21.8	21.8	22.8	24.1	25.0
	DBStd	2.15	1.13	0.99	1.09	1.26	1.49	1.47	1.65	1.63	1.56	1.38	1.26	1.33
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDD10.0	5056	486	459	499	446	422	373	365	365	354	398	424	465
	CDD18.3	2016	227	226	241	196	163	123	107	107	104	139	174	207
	CDH23.3	10591	1653	1916	1924	1122	633	271	207	179	182	387	821	1296
(13) Wind	WSAvg	3.0	2.9	3.1	2.7	3.2	3.0	2.8	3.2	3.3	3.1	3.0	3.0	3.0
	PrecAvg	1969	251	224	181	245	170	102	126	117	105	131	140	173
	PrecMax	2717	618	502	378	532	462	347	421	319	239	350	424	526
	PrecMin	1091	23	24	31	27	15	14	26	10	21	19	21	24
	PrecStd	449	147	115	105	138	111	79	91	88	61	83	99	116
	DB	29.9	31.1	30.3	29.1	28.1	27.6	27.1	27.0	27.2	28.0	30.0	29.8	29.8
	MCWB	25.7	24.5	26.0	25.4	24.3	23.0	22.5	22.6	22.6	22.7	24.6	25.1	25.1
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.0	30.0	29.6	28.3	27.3	26.2	26.1	25.7	26.1	26.9	28.3	28.9	28.9
	2% MCWB	25.3	25.2	25.6	25.0	23.8	22.8	22.7	22.3	22.6	22.4	23.7	24.9	24.9
	5% DB	28.4	29.3	29.0	27.6	26.7	25.6	25.3	25.0	25.2	26.1	27.4	28.1	28.1
	5% MCWB	24.9	25.1	25.4	24.5	23.3	22.4	22.1	21.9	21.8	22.1	23.3	24.3	24.3
	10% DB	27.9	28.7	28.4	27.0	26.0	24.9	24.6	24.3	24.4	25.3	26.6	27.4	27.4
	10% MCWB	24.6	25.0	25.0	24.0	22.8	21.7	21.5	21.5	21.3	21.7	22.9	23.9	23.9
	WB	27.3	27.1	27.3	26.6	25.4	24.8	24.2	23.9	24.1	24.5	25.6	26.1	26.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	29.3	29.1	29.4	28.1	27.2	26.1	25.8	25.4	25.9	26.4	28.1	28.3	28.3
	2% WB	26.2	26.3	26.5	25.9	24.7	24.0	23.6	23.4	23.4	23.8	24.9	25.6	25.6
	2% MCDB	27.9	28.3	28.7	27.4	26.5	25.8	25.3	24.9	25.2	25.6	27.2	27.8	27.8
	5% WB	25.6	25.8	26.0	25.4	24.2	23.2	22.9	22.8	22.8	23.2	24.3	25.1	25.1
	5% MCDB	27.5	27.9	28.2	27.1	25.9	25.1	24.6	24.4	24.5	25.1	26.5	27.3	27.3
	10% WB	25.2	25.5	25.6	24.7	23.7	22.5	22.3	22.3	22.1	22.6	23.7	24.6	24.6
	10% MCDB	27.1	27.7	27.7	26.5	25.4	24.4	24.0	23.9	24.0	24.6	25.8	26.7	26.7
(35) Mean Daily Temperature Range	MDBR	4.1	4.4	4.5	3.8	4.4	4.7	4.8	4.6	4.7	4.7	4.6	4.4	4.4
	5% DB MCDBR	4.8	5.3	5.2	4.5	4.9	5.4	5.7	5.5	5.4	5.6	6.0	5.3	5.3
	5% DB MCWBR	2.5	2.2	2.5	2.6	2.8	3.3	3.3	3.3	3.2	3.0	2.8	2.6	2.6
	5% WB MCDBR	4.3	4.1	4.5	3.8	4.1	4.4	4.3	4.2	4.8	4.6	5.0	4.7	4.7
	5% WB MCWBR	2.7	2.2	2.6	2.6	2.9	3.4	3.3	3.5	3.5	3.2	2.9	2.7	2.7
(40) Clear-Sky Solar Irradiance	taub	0.380	0.380	0.372	0.376	0.367	0.358	0.355	0.355	0.372	0.368	0.371	0.377	0.377
	taud	2.619	2.624	2.655	2.633	2.639	2.643	2.639	2.632	2.578	2.587	2.608	2.611	2.611
	Ebn at Noon	964	951	934	889	859	847	863	898	919	952	966	968	968
	Edh at Noon	103	100	93	89	82	79	81	87	99	103	103	104	104
(44) All-Sky Solar Radiation	RadAvg	6.41	6.14	5.54	4.64	3.76	3.49	3.71	4.41	5.29	6.00	6.34	6.36	6.36
	RadStd	0.51	0.50	0.43	0.28	0.30	0.25	0.26	0.26	0.31	0.42	0.48	0.53	0.53

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page